

Master's Thesis

submitted in partial fulfillment of the
requirements for the course "Applied Computer Science"

Bridging the Sim2Real Gap for Vision-Based Runway Detection: Weather Augmentation, Self-Training, and Diffusion-Based Style Transfer with YOLO Pose Models

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ISSN: 1612-6793

July 1, 2026

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Abstract

Vision-based runway detection on approach is a promising, infrastructure-independent building block for assisted and autonomous landing, but training data that covers diverse airports, viewpoints and, above all, adverse weather is scarce. Synthetic data offers a scalable, automatically labeled alternative, yet models trained on it suffer a *Sim2Real gap* when deployed on real imagery, especially under rain, glare, fog/snow and night conditions.

This thesis studies and closes that gap for a YOLOv8-pose model that detects the four runway corners, fine-tuned on nominal-condition synthetic data from LARD-V2 and evaluated on a weather-stratified real test set from LARD-V1. It first quantifies the gap through a controlled, shared-location cross-evaluation of five synthetic rendering sources and a combined source, identifying which source generalizes best overall and per weather scenario, and isolating a runway orientation/distance bias that confounds naive real-versus-synthetic comparisons.

It then evaluates three complementary techniques, ordered from the least to the most direct intervention on the domain shift: input-space weather augmentation with a ratio study; self-training with confidence-filtered pseudo-labels on unlabeled real landing sequences, extended with tracker-based temporal consistency; and label-preserving, diffusion-based style transfer (Cosmos Transfer 2.5), optionally compared against a CycleGAN baseline. All strategies are compared under one lightweight, reproducible protocol (a fine-tuned YOLOv8-pose model, a fixed 5000-image budget, and a weather-stratified real test set), reporting primarily mAP@[.50:.95] together with IoU-based localization and inference speed.

We find that the gap is only measurable once the test sets are matched on runway orientation as well as size, that it concentrates in adverse weather (fog being the hardest), and that self-training on unlabeled real footage is the most effective technique so far — improving adverse-weather splits by over ten mAP points — while weather augmentation helps only modestly and diffusion-based style transfer is ongoing.

Declaration on the use of AI in the context of examinations

In this work I have used AI as follows:

- How was AI used in the development of ideas?
 - e.g. where are open research questions, brainstorming, dialogue, ...
- How was AI used in working with literature and sources?
 - e.g. finding sources, semantic search, generate search strings
- How was AI used in the process of academic writing?
 - e.g. formatting of LaTeX, rephrasing, summaries, text generation, checking grammar
- How was AI used in the methodology and particularly in data collection and analysis?
 - e.g. finding relevant data points, data summary, generate python snippets for data analysis
- How was AI used in other activities?
 - e.g. emotional support, code generation, ...

I hereby declare that I have stated all uses completely.

Missing or incorrect information will be considered as an attempt to cheat.

Acknowledgements

thanks

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List of Listings

List of Abbreviations

YOLO	You Only Look Once
LARD	Landing Approach Runway Detection
UDA	Unsupervised Domain Adaptation
mAP	mean Average Precision
IoU	Intersection over Union
OKS	Object Keypoint Similarity

1 Introduction

1.1 Motivation

The final approach and landing is statistically the most accident-prone phase of flight. Guiding it today relies on ground infrastructure such as instrument landing systems and on satellite navigation, which are not available at every runway and can be degraded or spoofed. A forward-looking camera on the aircraft offers a low-cost, self-contained alternative: if the runway can be located in the image and its geometry recovered, the system can estimate the aircraft’s relative pose and support alignment during approach, reducing pilot workload and enabling assisted or autonomous landing [Fer+25].

Detecting the runway from the aircraft perspective is, however, a hard vision problem. The runway appears as a small, thin quadrilateral early in the approach and grows and shifts rapidly as the aircraft descends, so its *apparent size and distance* vary over orders of magnitude. Its *orientation and perspective* depend on the approach angle, glide slope and lateral offset, and the on-board *camera viewpoint* adds further variability. On top of this geometric variability, *weather and illumination* — rain, sun glare, fog, snow and night operations — degrade contrast and inject nuisances (streaks, halos, scattering, low light) precisely when reliable detection matters most. Runway-specific detectors such as YOLO-RWY [Li+24] and lightweight tracking-based landing-guidance pipelines [Fer+25; Aut26] illustrate both the practical interest in the task and its difficulty.

Framing the task as *keypoint (pose) estimation* of the four runway corners, rather than as plain bounding-box detection, is particularly attractive: the four corners directly parameterize the runway quadrilateral and can be fed into downstream pose-from-geometry estimators, while remaining cheap enough for real-time, on-board inference.

The central obstacle is training data. Annotated real approach imagery that spans a wide range of airports, runway geometries, approach angles and, crucially, weather and lighting conditions is scarce: real landing footage is costly to collect and its per-corner annotation is laborious, so publicly available labeled datasets are limited. Photorealistic simulators and geospatial rendering pipelines offer a way out — they synthesize large, automatically and exactly labeled datasets covering many locations and viewpoints [Duc+25] — and the Landing Approach Runway Detection (LARD) dataset family [Duc+23] exemplifies this approach. Its second iteration (LARD-V2) provides several independent synthetic image sources for the same set of runway locations.

1.2 Problem Statement

Models trained purely on synthetic imagery rarely reach their synthetic-domain performance once deployed on real images. This performance drop, commonly referred to as the *Sim2Real gap*, is a specific instance of *domain shift*: the statistical distribution of the training (source, synthetic) domain differs from that of the deployment (target, real) domain in texture, lighting, sensor noise, atmospheric effects and scene composition. For runway detection the gap is amplified by safety-critical operating conditions: the system must remain reliable precisely in the adverse weather and low-visibility scenarios (rain, glare, fog, snow, night) that are hardest to render faithfully and least represented in nominal synthetic data.

This thesis studies the Sim2Real gap for runway keypoint detection when a YOLOv8-pose model is fine-tuned on nominal-condition synthetic data from LARD-V2 and eval-

uated on a real, weather-annotated test set from LARD. It first *quantifies* the gap and then investigates three complementary families of techniques for *closing* it, ordered by how directly they intervene on the domain shift: input-space weather augmentation, self-training with pseudo-labels on unlabeled real footage, and generative style transfer that imports target-domain appearance onto the synthetic source.

Goal of the thesis. The goal of this thesis is to evaluate *which Sim2Real strategy best improves lightweight runway-corner detection from aircraft-view images when only limited annotated real data is available*. Concretely, the strategies are compared under a single, reproducible, lightweight setup (a fine-tuned YOLOv8-pose model, a fixed 5000-image training budget, and a shared weather-stratified real test set), so that observed differences are attributable to the adaptation strategy rather than to model capacity or training budget.

Research gap. Prior work tends to address these ingredients *separately*: some studies focus on generating and mixing synthetic runway data [Duc+25], others on runway-specific detector design [Li+24], weather-robust augmentation [Aut25], or domain adaptation [GL15]. In contrast, this thesis compares several Sim2Real strategies — weather augmentation, self-training, and diffusion-based style transfer — head-to-head under one lightweight, reproducible protocol, which makes their relative benefits directly comparable.

Label-preserving style transfer. A recurring difficulty for the generative strategy is that the four runway corners must remain geometrically valid after an image is restyled. Unpaired image-to-image translation with CycleGAN-style models can hallucinate or displace structure, which would silently corrupt the corner labels. This thesis therefore favors a *conditional diffusion* model that is guided by the source image’s structure so that it repaints appearance — lighting, texture, weather traits of the real target domain — while leaving the runway geometry, and hence the labels, in place; the CycleGAN route is kept only as a contrast that exposes the label-displacement risk.

1.3 Research Questions

The work is organized around four research questions (RQs) that follow a single logical thread: measure the gap, then progressively attack it. Each RQ states a *question*; the *experiment* that answers it is developed in the corresponding section (Sections 5 to 8), keeping the questions separate from the methodology.

- **RQ1 (Gap characterization).** How large is the Sim2Real gap for runway keypoint detection across the five LARD-V2 synthetic sources and a combined source, and which source generalizes best to the real test set overall and within each individual weather scenario? *Motivation:* the size and structure of the gap must be measured before any mitigation can be assessed, and the choice of synthetic source is itself a design decision reported to matter for runway detection [Duc+25].
- **RQ2 (Weather augmentation).** To what extent do input-space weather augmentations improve real-world generalization for a single synthetic source and for the combined source? Which augmentation ratios help and which harm, and can

augmentation compensate for a lack of dataset diversity? *Motivation:* augmentation is the cheapest, label-preserving robustness lever and is known to shift detector performance under adverse weather [Aut25], but its ratio and its interaction with dataset diversity are rarely quantified for this task.

- **RQ3 (Self-training).** Can self-training with confidence-filtered pseudo-labels on unlabeled real landing sequences reduce the gap, and does exploiting temporal consistency through an object tracker further improve the pseudo-labels? *Motivation:* unlabeled real landing footage is comparatively abundant, and self-training is a standard route to unsupervised domain adaptation [GL15]; tracking is already used in landing-guidance pipelines [Fer+25; Aut26] and can enforce temporal label consistency.
- **RQ4 (Style transfer).** Can diffusion-based style transfer import real-domain (and adverse-weather) appearance onto the synthetic source without breaking the geometric labels, and how does it compare to a GAN-based alternative that risks displacing keypoints? *Motivation:* style transfer attacks the appearance gap most directly, but only helps if the four-corner labels survive the transformation.

1.4 Contributions

This thesis makes the following contributions:

1. A **real-domain characterization of the Sim2Real gap** for runway keypoint detection, using a shared-location cross-evaluation across five synthetic rendering sources plus a combined source against the real LARD-V1 set. This complements the LARD-V2 benchmark (which studied cross-dataset generalization on *synthetic* test sets) and isolates the rendering source from the runway viewpoint.
2. A **vertical-angle test-set filtering** method and the finding that the gap is only measurable once the test sets are matched on runway *orientation* as well as size; without it, the real set spuriously appears easier than the synthetic sets. The residual synthetic difficulty is shown to be largely an image-quality effect concentrated on distant runways.
3. A **weather-stratified real benchmark**, extended with private landing sequences to compensate for the sparse adverse-weather representation, together with a style-feature analysis explaining why different sources suit different scenarios.
4. A systematic **study of weather augmentations** (glare, fog, rain, water refraction, night) and their mixing ratios, with a variance-controlled protocol, testing whether augmentation substitutes for dataset diversity.
5. A **self-training pipeline** on unlabeled real landing sequences that compares confidence thresholds, analyzes pseudo-label composition, and extends pseudo-labeling with a tracker-based temporal-consistency mechanism (ongoing).
6. An investigation of **control-guided diffusion style transfer** (Cosmos Transfer) for label-preserving domain translation of the synthetic source, with an optional CycleGAN comparison (ongoing).

1.5 Thesis Outline

Section 2 introduces the concepts required to follow the work: vision-based runway detection, YOLO pose estimation, the Sim2Real gap and domain-adaptation families, and the LARD datasets. Section 3 positions the thesis against related literature. Section 4 defines the shared experimental setup: datasets and weather splits, model, evaluation metrics and training protocol. Section 5 establishes and dissects the Sim2Real gap (RQ1). Section 6 studies weather augmentations (RQ2). Section 7 develops the self-training pipelines (RQ3). Section 8 investigates diffusion-based style transfer (RQ4). Section 9 compares the techniques and discusses limitations, and Section 10 concludes and outlines future work.

2 Background

This section introduces the concepts needed to follow the thesis. It is deliberately scoped to what the later experiments rely on: runway detection as a pose task, the YOLO family, the vocabulary of the Sim2Real gap and domain adaptation, and the LARD datasets.

2.1 Vision-Based Runway Detection as Pose Estimation

A runway projected into an approach image is well approximated by a planar quadrilateral. Detecting it can therefore be cast either as (i) bounding-box detection of the runway region, or (ii) localization of the four runway corners as *keypoints*. This thesis adopts the keypoint (pose) formulation: the model predicts a runway instance together with its four ordered corner points (e.g. near-left, near-right, far-left, far-right). The four corners are a compact, geometry-preserving representation that downstream modules can turn into a relative pose estimate through a perspective-n-point solve. A key practical consequence, relevant throughout the thesis, is that any label-altering transformation of the image (geometric augmentation, generative restyling) must keep the four corner annotations geometrically consistent.

2.2 The YOLO Family and YOLOv8-Pose

You Only Look Once (YOLO) (You Only Look Once) is a family of single-stage object detectors that predict objects densely in a single forward pass, trading a small accuracy margin for high inference speed suitable for onboard, real-time use. YOLOv8 extends the family with an anchor-free, decoupled detection head and a dedicated *pose* variant that adds a keypoint-regression branch, predicting a fixed number of keypoints per detected instance. For runway detection this branch is configured for four keypoints. All experiments in this thesis *fine-tune* a pretrained YOLOv8-pose checkpoint rather than training from scratch, so that the comparisons isolate the effect of the training *data* and adaptation technique rather than of the backbone initialization.

2.3 The Sim2Real Gap and Domain Shift

When a model is trained on a source distribution $\mathcal{D}_S$ (synthetic imagery) and deployed on a target distribution $\mathcal{D}_T$ (real imagery), differences between the two distributions cause

a drop in performance known as the *Sim2Real gap*. It manifests as *covariate shift* in low-level statistics (texture, color, sensor noise, compression), in mid-level appearance (lighting, shadows, atmospheric scattering) and in scene composition (which locations, viewpoints and object scales are represented). Adverse weather compounds the shift: rain streaks, water refraction on the lens, sun glare, fog/snow attenuation and night illumination are rarely modeled faithfully by nominal renderers and are under-represented in nominal synthetic data.

2.4 Families of Domain-Adaptation Techniques

The techniques studied in this thesis can be placed on a spectrum according to *where* they intervene on the shift:

- **Input-space augmentation.** Cheap, label-preserving transformations applied to the source images to synthesize target-like nuisances (e.g. weather). They do not require target data but only approximate real degradations.
- **Self-training / pseudo-labeling.** A form of unsupervised domain adaptation (Unsupervised Domain Adaptation (UDA)) that uses model predictions on unlabeled target data as training targets, optionally filtered by confidence and refined by temporal consistency. It uses real data directly but is sensitive to confirmation bias.
- **Generative style transfer.** Learns or applies a mapping that imports target appearance onto the source images while preserving content, so that labels remain valid. GAN-based mappings can distort geometry; diffusion-based conditional transfer aims to keep structure intact.

Sections 5 to 8 explore these families in this order, from the least to the most direct intervention on the gap.

2.5 The LARD Datasets

The Landing Approach Runway Detection (LARD) dataset family targets runway detection on approach. The first version (LARD-V1) provides, among others, a set of *real* approach images annotated with runway corners, which this thesis uses as the held-out real test set. The second version (LARD-V2) refines the operational design domain (a narrower approach cone, multi-runway airports) and, most importantly here, provides multiple *synthetic* image sources for a shared set of runway locations, generated through different rendering and imagery pipelines. This thesis uses five such sources — **flsim** (Flight Simulator) and **xplane** (X-Plane), which are full 3D simulators; **arcgis** and **bingmaps**, which are satellite-imagery based; and **ges** (Google Earth Studio), which blends satellite imagery with 3D rendering — plus a *combined* source that mixes all five. Because the sources share `location_ids`, images from different sources correspond to the exact same runway views, which is what makes the cross-source comparison in Section 5 a controlled experiment. The precise composition and the sampling scheme used to keep the training size fixed are detailed in Section 4.

3 Related Work

This section positions the thesis against four strands of literature that map onto its research questions: runway detection and the LARD benchmark, learning from synthetic data and the Sim2Real gap, weather-robust augmentation, and domain adaptation via self-training and generative style transfer. For each strand we make explicit how the cited work relates to, and differs from, this thesis.

3.1 Vision-Based Runway Detection and the LARD Benchmark

Runway detection for vision-based landing has been approached both as bounding-box detection and as geometry-aware localization. Runway-specific detectors such as YOLO-RWY [Li+24] train a dedicated YOLOv8 detection architecture on LARD-V1, report performance on the real nominal and edge-case test splits, and introduce a *data-augmentation module* spanning diverse weather conditions (rain, snow, fog, shadow, sun flare). Notably, the authors announce as future work the use of GANs for data augmentation and feature-enhancement modules to compensate for the limited night and adverse-weather data in LARD, as well as the extraction of runway *keypoints* for pose estimation — directions this thesis takes up directly. Beyond a single frame, landing-guidance systems combine detection with tracking and orientation estimation: Ferreira et al. [Fer+25] track the runway and estimate aircraft orientation across an approach, and runway/airside safety systems use detection-plus-tracking video analytics for surveillance [Aut26].

The LARD benchmark [Duc+23] provides both real and synthetic approach imagery annotated with runway corners. Its second iteration, LARD-V2 [Duc+25], addresses two limitations of LARD-V1 — an operational design domain (ODD) restricted to single-runway airports with an unrealistic approach cone, and few generation sources — by narrowing the approach cone, extending to multi-runway airports, proposing an extended-ODD metric (e-mAP) that accounts for both in-ODD and extended-ODD runways, and expanding source diversity to improve cross-domain generalization. The authors benchmark YOLOv11 detectors with mAP and e-mAP, training eleven models that differ only in training-data composition (each individual source, the full dataset, and five leave-one-out sets). Their findings are directly relevant here: the *full* configuration achieves the best average performance (every source contributes); the satellite-only sources ArcGIS and Bing Maps combine well with each other but less well with full 3D simulators; and *ges* attains the highest mAP among individual sources, likely because it blends satellite imagery with 3D rendering.

Relation to this thesis. Like YOLO-RWY, this thesis uses a lightweight YOLO detector, but formulates the task as four-corner *pose* estimation and focuses on *closing the Sim2Real gap* for a fixed detector rather than on architecture design. Whereas the LARD-V2 authors studied cross-*dataset* generalization with a single detector on synthetic test sets, this thesis repeats the analysis against the *real* LARD-V1 set, stratifies it into weather scenarios as weather-specific benchmarks (extended with private landing sequences to compensate for the sparse adverse-weather representation), and evaluates weather augmentation, self-training and style transfer as remedies. The strong YOLO benchmark and the object-detection findings, which the LARD-V2 authors relate to the datasets’ rendering foundations, transfer naturally to the pose-estimation setting adopted here. Real-time detector alternatives such as RT-DETR [Zha+24] are noted as possible baselines but are outside the scope, which fixes the detector to isolate the adaptation

strategy.

3.2 Learning from Synthetic Data and the Sim2Real Gap

Synthetic imagery from simulators and geospatial renderers enables large, exactly labeled training sets, and domain randomization [Tob+17] broadens the training distribution to improve transfer. For runways specifically, LARD-V2 generates imagery from five pipelines (flight simulator, ArcGIS, Bing Maps, Google Earth Studio, X-Plane) and shows that mixing sources improves cross-domain generalization [Duc+25]. These works establish that synthetic data is viable but that a residual Sim2Real gap remains, especially away from nominal conditions.

Relation to this thesis. This thesis first *quantifies* that residual gap in a controlled, shared-location cross-source comparison against the real domain (Section 5). Its methodological contribution is to show that the gap is only measurable after controlling for a runway *orientation* bias (vertical angle) in addition to runway size — without which the real set spuriously appears easier than the synthetic sets.

3.3 Weather-Robust Augmentation

Detector accuracy degrades markedly under rain, fog, glare and low light, and library-based augmentation such as **albu**mentations [Bus+20] provides fast, composable weather-like transformations; robustness studies quantify how much such degradations move detection metrics [Aut25]. Closest to this thesis, Rüter et al. [RDD25] systematically investigate how individual image augmentations affect the sim-to-real generalization of perception models, reporting per-augmentation performance changes relative to a baseline and highlighting the combinatorial explosion of augmentation configurations.

Relation to this thesis. Section 6 applies weather augmentations to the runway-pose task and, going beyond an on/off comparison, sweeps the augmentation *ratio* per weather scenario, controls the dominant sampling variance by fixing the training set, and tests whether appearance augmentation can substitute for the diversity of a multi-source training set.

3.4 Self-Training, Pseudo-Labeling and Temporal Consistency

Unsupervised domain adaptation aligns a model to an unlabeled target domain; classic approaches range from feature alignment by backpropagation [GL15] to self-training, where confident model predictions on target data serve as pseudo-labels [Lee13], typically in a student–teacher setup combining pseudo-label and consistency objectives. In video, object trackers can propagate labels across frames to enforce temporal consistency and recover objects the per-frame detector misses [Fer+25; Aut26; Ye+22].

Relation to this thesis. Section 7 instantiates confidence-filtered self-training on unlabeled real landing sequences, compares thresholds, analyzes the pseudo-label composition across iterations, and extends the loop with a tracker (OSTrack) that propagates corner pseudo-labels forward and backward along a sequence to improve their completeness for distant runways.

3.5 Generative Style Transfer for Domain Adaptation

Image-to-image translation adapts source images to look like the target domain. CycleGAN [Zhu+17] popularized unpaired translation for Sim2Real, but its lack of explicit geometric constraints can hallucinate or displace structure and thus break keypoint labels. Conditional diffusion models with a control branch, such as the Cosmos platform [NVI25], can be guided by the source structure (edge or segmentation maps, reference images, prompts) to repaint appearance while preserving scene geometry.

Relation to this thesis. Section 8 uses control-guided, structure-preserving diffusion to import real and adverse-weather appearance onto the combined synthetic source *while keeping the four-corner labels valid*, and optionally contrasts it with a CycleGAN baseline to quantify the label-displacement risk.

4 Experimental Setup

This section defines the elements shared by all experiments: the datasets, the comparability-preserving test-set filtering, the model and training protocol, and the evaluation metrics. Experiment-specific choices are described in the respective sections (Sections 5 to 8).

4.1 Datasets

Synthetic training sources (LARD-V2). Training uses the five synthetic sources of LARD-V2 — `flsim` (Flight Simulator), `arcgis` (ArcGIS), `bingmaps` (Bing Maps), `ges` (Google Earth Studio) and `xplane` (X-Plane) — generated for approximately **250 airports worldwide**. The full synthetic training pool comprises **66 565 images**. To make the sources directly comparable, the images are grouped so that every runway viewpoint is represented by the same number of images in each source; this grouping reduces the pool to **7 715 images per source (38 575 in total)**. A key consequence of this design is that the individual sources share `location_ids`: image i in each source depicts the *same* runway view rendered by a different pipeline, which turns the cross-source comparison of Section 5 into a controlled experiment in which the rendering source is the only variable.

Training budget and the combined source. From this pool, each training run fine-tunes on a fixed budget of **5000 images**. The `combined` source (also referred to as `all`) mixes all five pipelines in equal proportion at matched locations; to respect the 5000-image budget it therefore covers roughly *one fifth* as many distinct locations as an individual source (each location contributing five renderings instead of one). This sets up an explicit trade-off between *rendering-source diversity* and *runway-viewpoint coverage* that recurs throughout the thesis. An extended variant that relaxes the budget, `all_extended`, is included where noted.

Real test set (LARD-V1) and weather splits. Real-world performance is evaluated on a held-out set of **1880 real images** from LARD-V1, composed of **1557 nominal** images and **323 edge cases**. The edge cases, together with the nominal split, are stratified into five weather/lighting scenarios: *nominal*, *glare*, *rain*, *fog* and *night*. Because the nominal scenario accounts for almost three quarters of the set, the *overall* real metric is

nominal-dominated; per-scenario numbers are therefore used whenever adverse conditions are discussed. To enable a fair comparison against the synthetic domain, an equally sized sample is drawn from the synthetic sources (from images not used for training). Table 1 summarizes the composition.

Table 1: Real test set (LARD-V1), composition and weather stratification. The nominal split dominates, which is why per-scenario metrics are reported for adverse conditions.

Split	# Images
Nominal	1557
Edge cases (glare + rain + fog + night)	323
Total	1880

Unlabeled real data and the custom test set (self-training). The self-training experiments (Section 7) additionally use *unlabeled* real landing sequences collected from public YouTube channels, captured at airports and channels that are *disjoint* from the LARD-V1 test set. The sequences span the same adverse-weather scenarios as the test splits (nominal, glare, rain, fog) and are decomposed into approximately **1800 frames**. A subset of these sequences forms a *custom* landing-sequence test set (with glare, fog and rain splits) used to complement the LARD-V1 evaluation.

4.2 Test-Set Filtering for Comparability

A naive comparison between the synthetic and real test sets is confounded by differences in the *apparent runway size* and *orientation* of their runways. This thesis therefore filters the test sets along two axes before comparing them:

- **Runway size.** The apparent runway size is measured as the $\log_{10}$ of the fraction of image pixels occupied by the runway, and the test sets are filtered to a common size distribution.
- **Vertical angle.** Runways in the synthetic test sets frequently appear at oblique orientations, whereas real approaches are mostly aligned vertically. The *vertical angle* is defined as the absolute value of the angle of the runway with respect to the image y -axis, taking values in $[0^\circ, 90^\circ]$; it measures the runway’s “skewness” independently of direction. Matching this distribution across test sets removes an orientation bias that otherwise flatters the real set.

The effect of adding the vertical-angle criterion on top of the size criterion is analyzed in Section 5.

4.3 Model

All experiments fine-tune a pretrained **YOLOv8-pose** model with four keypoints corresponding to the runway corners. Fine-tuning from a pretrained checkpoint (rather than training from scratch) keeps the setup lightweight and reproducible and ensures that differences in results are attributable to the training data and adaptation technique rather

than to backbone initialization. This choice is further motivated by the LARD-V2 benchmark, on which the YOLO family performs strongly (Section 3.1).

State the exact variant (n/s/m/l), input resolution, optimizer, learning-rate schedule, batch size and hardware.

4.4 Training Protocol

Two training budgets are used, depending on the experiment:

- **Full budget (default).** 200 epochs with early-stopping *patience* of 25 epochs. Used for the Sim2Real gap characterization (Section 5), self-training (Section 7) and style transfer (Section 8).
- **Reduced budget (weather-augmentation sweep).** 20 epochs, used only for the weather-augmentation study (Section 6) to keep the large augmentation $\times$ ratio sweep tractable.

Sampling variance and repeated runs. Two sources of randomness strongly affect the mAP of a single run: (i) the specific images sampled into the fixed 5000-image budget (especially for the **combined** source, which re-samples locations), and (ii) the specific images selected to receive an augmentation. To control for this, each configuration in Section 6 is trained with **4 independent runs**. As detailed there, the **combined** training set is additionally *fixed* (sampled four times, best retained) so that the reported variance is attributable to the augmentation alone.

4.5 Evaluation Metrics

The **primary metric** is keypoint mean Average Precision (mAP) averaged over the ten Object Keypoint Similarity (OKS)/Intersection over Union (IoU) thresholds from 0.50 to 0.95 in steps of 0.05, denoted **mAP@[.50:.95]** (equivalently mAP50–95). It is the strictest single-number summary and rewards accurate, well-localized runway quadrilaterals rather than mere presence. mAP@0.50 is reported alongside it as a looser reference. Where a single localization figure is more informative — for example the runway-size analyses of Section 5 — the **IoU** between the predicted and ground-truth runway region is reported directly. Pixel-level precision/recall of the runway corners and **inference speed** (throughput, frames per second on the reference hardware) are reported as secondary, deployment-oriented metrics, in line with the edge-deployable target application.

All metrics are reported *overall* (entire real test set) and *per weather scenario*; because the overall figure is nominal-dominated, per-scenario numbers are used when reasoning about adverse conditions.

What counts as a meaningful improvement. Given the four-run variance-control protocol, an adaptation strategy is considered to *meaningfully* improve over the synthetic-only baseline on a given split when its mean mAP@[.50:.95] gain exceeds the run-to-run spread of the baseline. As a practical guide, single-point gains (~ 1 mAP) are treated as marginal, whereas gains of several mAP points on the adverse-weather splits — where the gap is largest — are treated as substantive.

Fix the OKS keypoint scales for the 4-corner runway pose, the detection confidence/IoU thresholds, and the exact spread value used for the “meaningful improvement” criterion once baseline variance is measured.

5 Establishing the Sim2Real Gap

This section addresses **RQ1**. It quantifies the Sim2Real gap for runway keypoint detection, identifies which synthetic source generalizes best to real data overall and per weather scenario, and — as a central methodological finding — shows that a fair measurement of the gap requires controlling for an *orientation* bias between the synthetic and real test sets, not only for runway size. The analysis proceeds from a first, counter-intuitive observation (the real set appears *easier* than the synthetic sets) to its resolution and to a weather-stratified view that motivates the remainder of the thesis.

5.1 Experimental Design

Seven models are fine-tuned under the full budget (200 epochs, patience 25; Section 4.4): one per individual synthetic source (`flsim`, `arcgis`, `bingmaps`, `ges`, `xplane`), the `combined/all` source, and the extended `all_extended` variant. Each model is evaluated on the test set of every synthetic source and on the real LARD-V1 set (as a whole and per weather scenario). Since the sources share `location_ids` (Section 4.1), the cross-source matrix isolates the rendering pipeline from the runway viewpoint.

5.2 Cross-Source Evaluation: A First, Counter-Intuitive Result

Table 2 reports the cross-source matrix when the test sets are matched on runway size only. Two patterns emerge. First, among the *synthetic* test sets, each model performs best on its own source (the diagonal dominates), the expected signature of a domain effect. Second, and counter-intuitively, the *real* test set yields the *best* score for almost every training source — e.g. `ges` reaches 0.653 on real versus 0.579 on its own test set. Taken at face value this *contradicts* the Sim2Real hypothesis, since the models generalize to real data better than to held-out synthetic data.

Table 2: Cross-source evaluation, mAP@[.50:.95], test sets matched on runway *size* only. Rows: training source; columns: evaluation test set. The real column is the best or near-best entry in every row.

Train \ Test	flsim	arcgis	bingmaps	ges	xplane	real
flsim	0.634	0.387	0.393	0.424	0.434	0.580
arcgis	0.438	0.558	0.533	0.454	0.436	0.610
bingmaps	0.434	0.564	0.541	0.484	0.409	0.617
ges	0.498	0.450	0.435	0.579	0.468	0.653
xplane	0.400	0.389	0.402	0.417	0.520	0.555

5.3 Resolving the Contradiction: The Vertical-Angle Bias

Investigating the cause revealed a distribution mismatch that the size-only filtering does not remove: runways in the synthetic test sets frequently appear at oblique orientations, whereas the real approaches are mostly aligned vertically. Because off-axis runways are harder, the synthetic sets were effectively *harder* than the real set, inflating the apparent real performance. Following the definition in Section 4.2, the test sets are re-filtered to match *both* the runway-size and the vertical-angle distributions.

Table 3 reports the resulting matrix. After matching orientation, the real set is *no longer* the best column: each model now generalizes best to its own synthetic source (e.g. **flsim** 0.673 on itself versus 0.557 on real; **ges** 0.644 versus 0.638 on real). This *enables* the Sim2Real hypothesis — the models do generalize worse to real data than to their training source. However, the differences are small, so this is best described as a *modest* rather than a dramatic gap.

Table 3: Cross-source evaluation, mAP@[.50:.95], test sets matched on runway *size and vertical angle*. After controlling for orientation, the diagonal (in-source) entry is the best in every row and the real column is no longer the maximum.

Train \ Test	flsim	arcgis	bingmaps	ges	xplane	real
flsim	0.673	0.416	0.441	0.491	0.480	0.557
arcgis	0.497	0.626	0.601	0.517	0.478	0.593
bingmaps	0.483	0.622	0.612	0.537	0.449	0.599
ges	0.565	0.496	0.480	0.644	0.504	0.638
xplane	0.432	0.450	0.467	0.471	0.557	0.543

5.4 Runway-Size (Distance) Analysis

Breaking performance down by apparent runway size clarifies *where* the gap lives. For a model trained on **flsim** and evaluated on its own source versus on real data, the two curves are nearly equal for small (distant) runways and diverge in favour of the synthetic source only for larger (nearer) runways. Adding the vertical-angle criterion shifts the cross-over point from a runway size of -2.87 to -3.37 (a difference of 0.5 on the $\log_{10}$ scale) and shrinks the difference at larger runways: the gap thus becomes *larger* and is felt at *smaller* runways than before. For the very smallest runways, the real set still occasionally outperforms the synthetic source. The same pattern holds when other synthetic sources replace **flsim**.

Figure placeholder — insert plot: Runway-size vs. IoU, model trained on flsim, evaluated on flsim vs. real, before and after adding the vertical-angle filter (cross-over at -2.87 vs. -3.37).

5.5 Why Synthetic Data Is Harder: Image-Quality Causes

That sim-trained models generalize surprisingly well — and sometimes better — to real data than to their own synthetic source, especially relative to *other* synthetic sources, points to a difference in *image quality* rather than a small domain gap. The effect is most pronounced for distant runways and makes runway detection on synthetic data genuinely harder. Comparing synthetic and real images at matched runway size reveals three recurring causes:

- **Runway/background separability.** On synthetic imagery, distant runways are often hard to distinguish from their pixel surroundings, leading to missed or poorly localized detections relative to equally distant real runways (which are rendered/captured at higher effective resolution).
- **Confusion with other instances.** Runways are frequently confused with parallel taxiways/roads or with neighbouring runways. This happens on both domains but appears more often on synthetic data.
- **Edge sharpness.** Runway edges tend to be sharper in real images than in synthetic ones, which yields higher precision and, almost always, a higher IoU.

A further, systematic error is independent of image quality: in *near* perspectives the model consistently under-covers the runway threshold markings (the “piano keys”), predicting only the near portion of the quadrilateral. Other mismatches — time of day, lighting contrast and runway appearance — also contribute to isolated failures.

Figure placeholder — insert plot: Qualitative examples at matched runway size: synthetic vs. real, ground truth (green) and YOLOv8-pose prediction (blue), with per-image Prec/Rec/IoU (separability, instance confusion, edge sharpness).

5.6 Weather-Stratified Analysis

Splitting the (size- and angle-filtered) real set into weather scenarios shows that the overall real performance is carried by the dominant nominal split and hides a large spread across conditions. Table 4 reports mAP@[.50:.95] per scenario. Three observations stand out:

- **Performance depends strongly on the scenario**, and *fog* is by far the hardest (≈ 0.21 – 0.32 across sources).
- **The best source is scenario-dependent.** The satellite-imagery sources (*ges*, *arcgis*, *bingmaps*) are strongest under *glare* and *nominal*, whereas the full 3D simulator *flsim* is strongest under *rain*, *fog* and *night*.
- **The combined source generalizes best overall.** *all/all_extended* is best or near-best in almost every scenario (e.g. nominal 0.636, night 0.524), confirming that *rendering-source diversity outweighs runway-viewpoint coverage* in this setup, even though *combined* covers only about one fifth as many locations.

Table 4: Real test set per weather scenario, mAP@[.50:.95] (size- and angle-filtered), by training source. `combined = all`. Fog is the hardest scenario; the best source varies by scenario.

Train source	Nominal	Glare	Rain	Fog	Night
flsim	0.533	0.443	0.596	0.315	0.376
arcgis	0.586	0.507	0.400	0.224	0.268
bingmaps	0.589	0.539	0.425	0.209	0.278
ges	0.628	0.538	0.460	0.210	0.262
xplane	0.533	0.450	0.286	0.221	0.282
all	0.636	0.586	0.527	0.280	0.406
all_extended	0.623	0.570	0.603	0.315	0.524

The good aggregate real performance is therefore largely attributable to the nominal split’s dominance; the adverse splits reveal the domain shift that the remaining techniques target.

To understand *why* different sources suit different scenarios, the datasets are characterized in a style-oriented feature space, using (i) a small interpretable descriptor (brightness, contrast, saturation, sharpness as the variance of the Laplacian, edge density, colorfulness) and (ii) a texture descriptor given by the averaged Gram matrices of a VGG16 network pretrained on ImageNet-1k. A UMAP projection of these features separates the sources into clusters and places the real domain apart from all synthetic sources, confirming a genuine style gap and motivating the appearance-level interventions of Sections 6 and 8.

Figure placeholder — insert plot: UMAP of style-oriented image features (interpretable + VGG-Gram) for the five synthetic sources and the real domain, showing a distinct “real” cluster.

5.7 Discussion

The gap is real but modest once orientation is controlled for, and the surprisingly strong real-domain performance is largely explained by higher real image quality rather than by the absence of a domain shift. Two takeaways are worth carrying forward. First, cross-domain comparisons for runway detection *must* control for runway size *and* orientation, or they will systematically flatter the real set. Second, the gap concentrates in adverse weather — especially fog, rain and night — and no single synthetic source dominates all scenarios, although the combined source comes closest. From here on, the techniques target **generalization to real adverse-weather conditions** when training on nominal-condition synthetic data.

6 Weather Augmentations

This section addresses **RQ2**: whether input-space weather augmentations improve real-world generalization, which mixing ratios help or harm, and whether augmentation can

compensate for a lack of synthetic diversity. It is the least direct intervention on the domain shift — it approximates target-domain nuisances without using any real data.

6.1 Augmentations and Training Sources

Weather augmentations are drawn from the `albumentations` library and applied *individually* to synthesize the adverse conditions of the real test set: **glare** (RandomSunFlare), **fog** (RandomFog), **rain** (RandomRain), **water refraction** (WaterRefraction) and **night** (a darkening transform). A qualitative comparison of these augmentations against real glare/fog/rain images confirms that they approximate, but do not perfectly reproduce, the real degradations.

Following Section 5, augmentations are applied to the two most relevant training sources: **ges** (best individual source on the real set) and **combined** (best overall). Evaluation is per weather scenario on the LARD test set. This pairing makes it possible to test whether augmentation benefits depend on the underlying dataset diversity.

6.2 Controlling Sampling Variance

A first sweep exposed a methodological problem: the run-to-run mAP variance was so large that it masked any augmentation effect. For the **combined** source, an augmentation sweep on the rain split showed *no* improvement over the no-augmentation baseline and a wide spread across the four runs; for **ges**, the 0%-ratio runs had no spread at all (the training set is fixed), confirming that the variance originates from two independent sampling steps: (i) re-sampling the **combined** training set and (ii) sampling which images are augmented.

To attribute variance to the augmentation alone, the protocol is tightened (Section 4.4):

- the **combined** training set is *fixed* by sampling it four times and keeping the highest-performing draw;
- for each augmentation ratio, four independent runs are performed and the *maximum* achieved mAP is reported, i.e. an estimate of the best attainable performance at that ratio.

The ratios evaluated are **0.15, 0.30, 0.45, 0.60, 0.75**. This choice reflects the combinatorial blow-up of a full grid: as noted in the augmentation-influence literature [RDD25], sweeping augmentations, ratios and sources jointly yields on the order of 10^9 configurations, so the search is deliberately scoped and, as a bridge to the later methods, the data/augmentation mix is tuned rather than exhaustively searched.

6.3 Results

Table 5 summarizes, per scenario, the fixed baselines for **ges** and **combined** together with the best- and worst-performing augmentation×ratio found. The analysis answers the three sub-questions of RQ2.

Do augmentations behave as expected? Not straightforwardly. The augmentation that matches a scenario is *not* generally the best augmentation for that scenario — e.g. on the glare split the best result for **ges** comes from water *refraction*, not glare, and on several splits the scenario-matched augmentation is among the *worst*. Weather robustness thus does not decompose cleanly into “augment with the target weather”.

Which ratio helps and which harms? The effect is strongly ratio-dependent and non-monotonic. Two representative cases on **ges**: under water *refraction* on the rain split, raising the ratio from 0.60 to 0.75 improves mAP by almost 10 points; conversely, *glare* augmentation on the fog split yields both the best and the worst outcomes, dropping by more than 10 mAP points when the ratio is raised from 0.45 to 0.75. There is no single universally safe ratio; overly high ratios tend to over-corrupt the source.

Do augmentations compensate for missing diversity? Only weakly. For the low-diversity **ges** source, the best augmentation can improve a scenario substantially, but it does *not* reach the performance of the **combined** baseline. And when the diversity is already high (**combined**), augmentation helps only very little — the **combined** baseline is consistently at or above the best single augmentation applied to it. Appearance augmentation therefore does not substitute for rendering-source diversity; at best it narrows the gap for a single-source model.

Table 5: Weather augmentation on **ges** and **combined**, per scenario: fixed baseline and best/worst augmentation×ratio (max over 4 runs, reduced 20-epoch budget). The scenario-matched augmentation is rarely the best, and the **combined** baseline is hard to beat.

Scenario	Source	Baseline	Best / Worst augmentation (ratio)
Glare	ges	–	refraction 15% / glare 75%
Glare	combined	–	refraction 30% / glare 75%
Rain	ges	–	dark 60% / glare 75%
Rain	combined	–	fog 15% / fog 60%
Fog	ges	–	glare 45% / rain 75%
Fog	combined	–	dark 15% / fog 75%
Night	ges	–	rain 30% / glare 75%
Night	combined	–	dark 30% / glare 45%

Fill baseline mAP values from the fixed-dataset runs and, if desired, the best/worst mAP alongside each configuration.

Figure placeholder — insert plot: GES/combined baseline with best and worst augmentation per scenario (min–max bands); and the per-augmentation ratio curves for the ges rain and fog splits (refraction +10 mAP at 0.75; glare drop > 10 mAP 0.45→0.75).

6.4 Discussion

Weather augmentation is a cheap, label-preserving baseline that requires no real data, but its benefit here is limited and unstable: the scenario-matched augmentation is often not the best, the useful ratio is condition-specific, and augmentation does not recover the advantage of source diversity. Part of this instability is methodological — four runs are not always enough to average out the sampling variance, and more runs are expensive — which itself is a finding: robust augmentation studies on this task need either many runs or a fixed, controlled data mix. These limitations motivate the data-driven techniques of Section 7 (which uses real, unlabeled data) and Section 8 (which imports real appearance generatively).

7 Self-Training with Pseudo-Labels

This section addresses **RQ3**: it tackles the Sim2Real gap directly through UDA by self-training on *unlabeled real* landing sequences. A confidence-filtered self-training loop is developed and evaluated first; a tracker-augmented extension that exploits temporal consistency is then introduced and is *ongoing*.

7.1 Unlabeled Real Data

The target-domain data are the unlabeled real landing sequences described in Section 4.1: public YouTube footage from airports and channels *disjoint* from the LARD-V1 test set, spanning the nominal, glare, rain and fog scenarios and decomposed into approximately 1800 frames. A subset forms the custom landing-sequence test set (glare, fog, rain) used alongside the LARD-V1 evaluation. Keeping the sources disjoint from the test set avoids leakage and tests genuine generalization.

7.2 Confidence-Filtered Self-Training Loop

The loop (Figure 1) runs for **4 iterations**; each iteration fine-tunes a fresh YOLOv8-pose model for 200 epochs.

1. **Iteration 0** trains on the synthetic **combined** dataset only (equal numbers of images from the five sources).
2. The trained model predicts runway pose on the unlabeled real frames, each prediction carrying a confidence score.
3. Predictions with confidence below a threshold τ are discarded; the retained m (image, pseudo-label) pairs are moved into the training set and an equal number of synthetic images is removed, keeping the training size fixed at 5000 ($5000 - m$ synthetic + m pseudo-labeled).
4. The model is retrained on the updated set and the process repeats.

Two thresholds are compared, $\tau = 0.65$ and $\tau = 0.8$, trading pseudo-label *quantity* against *quality*.

*Figure placeholder — insert plot: Self-training loop: LARD-V2 synthetic pool
→ train-set (5k) → YOLOv8-pose → predictions+confidence on unlabeled
YouTube frames → confidence > τ filter → pseudo-labeled data back into
train-set; evaluated on the LARD test set and the custom landing-sequence
test set.*

Figure 1: Confidence-filtered self-training pipeline (4 iterations, thresholds $\tau \in \{0.65, 0.8\}$).

7.3 Results on the LARD-V1 Test Set

The two thresholds yield similar results on the LARD test set. On the nominal split the gain over the iteration-0 baseline is small, but on the adverse splits self-training improves $\text{mAP}@[.50:.95]$ by more than **10 points**, with most of the improvement realized already at the first iteration (Figure 2). Self-training on unlabeled real data thus transfers real-domain appearance far more effectively than weather augmentation did in Section 6.

*Figure placeholder — insert plot: $\text{mAP}@[.50:.95]$ by iteration (0–3) for the
LARD test set, per scenario (nominal, fog, rain, glare, night), thresholds 0.65
vs. 0.8.*

Figure 2: Self-training on the LARD test set: large gains on adverse splits, marginal on nominal; thresholds behave similarly.

7.4 Results on the Custom Landing-Sequence Test Set

On the custom landing-sequence test set the gains are considerably larger, and the threshold matters more — particularly for *fog*. Overall the baseline is exceeded far more than on the LARD test set, by up to **35 mAP points** on the glare split (Figure 3). This is expected: the custom sequences come from the same distribution as the pseudo-labels, so self-training adapts to them most strongly.

7.5 Pseudo-Label Composition

Analyzing which pseudo-labels enter the training set (at $\tau = 0.8$) reveals a subtlety of self-training. The *rain* pseudo-labels are strongly represented at iterations 1 and 2 (both in training-set proportion and acceptance ratio), yet the best rain-split performance is a modest +7 mAP reached already by the iteration-1 model. In other words, the iteration-0 predictions provided the most *useful* rain pseudo-labels, even though later iterations accumulate many more of them — a reminder that pseudo-label *quantity* does not equal quality and that self-training is exposed to confirmation bias.

Figure placeholder — insert plot: $mAP@[.50:.95]$ by iteration for the custom test set (fog, rain, glare), thresholds 0.65 vs. 0.8; up to +35 mAP on glare.

Figure 3: Self-training on the custom landing-sequence test set: larger gains and stronger threshold dependence than on LARD.

Figure placeholder — insert plot: Per-scenario pseudo-label train-set proportion and acceptance ratio by iteration ($\tau = 0.8$): rain dominates iterations 1–2.

7.6 Tracker-Augmented Temporal Pseudo-Labels (Ongoing)

So far the pseudo-labels use the per-frame YOLOv8-pose predictions without any additional filtering. Because the data are *video* landing sequences, combining the detector with an object tracker can exploit *temporal consistency*: a high-confidence detection can be propagated *forward and backward* through the sequence to recover the runway in frames where the detector alone is uncertain (e.g. distant or transiently degraded). This can be interpreted as distilling a pre-trained tracker into temporally consistent 4-corner pseudo-labels. A first qualitative comparison of YOLOv8-pose against YOLOv8-pose + OSTRacker [Ye+22] shows the tracker recovering runways that the single-frame detector misses (Figure 4).

Figure placeholder — insert plot: Qualitative comparison: YOLOv8-pose vs. YOLOv8-pose + OSTRacker on landing-sequence frames; the tracker recovers the runway in low-visibility frames.

Figure 4: Tracker-augmented pseudo-labeling (ongoing): temporal propagation recovers detections missed per-frame.

This extension is *in progress*. The plan is to regenerate the pseudo-labels with tracker-based temporal consistency and repeat the self-training loop for both thresholds, comparing against the confidence-only variant above.

Once complete: report tracker-augmented self-training results per scenario for both thresholds, and quantify the benefit of temporal consistency, especially for distant runways.

7.7 Discussion

Self-training uses real data directly and therefore attacks the domain shift more effectively than augmentation, delivering double-digit mAP gains on the adverse splits and up to +35 mAP on in-distribution landing sequences, at the cost of sensitivity to pseudo-label noise (as the rain-composition analysis shows). The threshold choice is nearly neutral on LARD but matters on the custom set. The tracker-augmented variant is expected to improve pseudo-label completeness in exactly the hard, distant-runway regime identified in Sections 5.4 and 5.5.

8 Diffusion-Based Style Transfer

This section addresses **RQ4**: it targets the domain and Sim2Real gap by transferring concrete style components of the target domain onto the labeled synthetic source — the **combined** dataset — thereby injecting domain information. It is the most direct appearance-level intervention: rather than approximating weather with library effects (Section 6) or relying on the model’s own predictions (Section 7), it repaints the source images to look like the target. This stage is *planned/ongoing*.

8.1 Motivation and Label Preservation

Style transfer imports stylistic properties of the real domain — textures, colour contrasts — and weather/time-of-day scenarios that either look different or are absent in the synthetic data. The style gap quantified in Section 5.6 (the distinct “real” cluster in the UMAP of style features) is precisely what this stage aims to close. The core constraint is *label preservation*: the four runway corners must remain geometrically valid after restyling. CycleGAN-style unpaired translation can change texture and colour but is hard to control, and if the transformation displaces the four corners the synthetic labels become incorrect.

8.2 Method: Control-Guided Diffusion (Cosmos Transfer)

Recent diffusion approaches use a *control branch* that steers the diffusion process. This branch is an additional model that combines the source image with *control modalities* such as text prompts, reference images or segmentation/edge maps. By conditioning generation on the source structure, the original labels are preserved far better than with GAN-based approaches. The thesis uses the open-source **Cosmos Transfer** diffusion model, conditioned on the structure of the **combined** synthetic images so that real and adverse-weather appearance is applied while the runway geometry — and hence the 4-corner labels — is retained. The restyled images are then used to fine-tune YOLOv8-pose under the full budget.

Fix the exact Cosmos Transfer version and the control modality used (edge/segmentation/depth), the target styles/weather applied, prompt settings, and the number of restyled images.

Table 6: Planned evaluation: control-guided diffusion style transfer applied to **combined**, on the real test set per weather scenario, relative to the **combined** baseline.

Configuration	Nominal	Glare	Rain	Fog	Night	Overall
combined (baseline)	–	–	–	–	–	–
+ Cosmos Transfer	–	–	–	–	–	–

Figure placeholder — insert plot: Style-transfer examples: synthetic source, edge/segmentation control map, and diffusion-restyled output with preserved runway geometry.

8.3 Optional Comparison: CycleGAN Baseline

Time permitting, the diffusion results are compared against a CycleGAN baseline that applies the same source→target translation but may displace the corners. The comparison quantifies both the downstream accuracy and the label-integrity advantage of the control-guided diffusion approach (the corner displacement introduced by the translation itself).

8.4 Discussion

Control-guided diffusion aims to combine the strengths of the earlier techniques: like augmentation it keeps labels exact and needs no target labels, and like self-training it grounds the adaptation in real target appearance — while, unlike CycleGAN, it is steerable and structure-preserving. Its main costs are compute and the fidelity and controllability of the generative model. The cross-technique comparison in Section 9 will place these results alongside weather augmentation and self-training once available.

9 Discussion

This section draws the experiments together and compares the techniques as answers to a single question: how best to close the Sim2Real gap for runway keypoint detection under adverse weather when training on nominal synthetic data.

9.1 Comparing the Techniques

The three intervention families differ sharply in effectiveness on this task. *Weather augmentation* (Section 6) is cheap and label-preserving but delivers only limited, unstable gains: the scenario-matched augmentation is often not the best, the useful ratio is condition-specific (e.g. water refraction +10 mAP on the **ges** rain split at ratio 0.75, but glare augmentation dropping > 10 mAP on the fog split from 0.45 to 0.75), and it does not recover the advantage of source diversity. *Self-training* (Section 7) is markedly stronger: using unlabeled real footage it improves the adverse LARD splits by more than 10 mAP and the in-distribution landing sequences by up to 35 mAP, with the confidence threshold

nearly neutral on LARD. *Style transfer* (Section 8) is expected to complement these but is still ongoing. Table 7 consolidates the best configuration of each available technique against the **combined** baseline.

Table 7: Best configuration of each technique vs. the **combined** baseline, real test set per weather scenario and overall (mAP@[.50:.95]). Style transfer is pending.

Technique	Nominal	Glare	Rain	Fog	Night	Overall
combined (baseline)	0.636	0.586	0.527	0.280	0.406	–
+ Weather augmentation (best)	–	–	–	–	–	–
+ Self-training (best)	–	–	–	–	–	–
+ Style transfer (Cosmos)	–	–	–	–	–	–

Fill the best-configuration rows once the augmentation, self-training and style-transfer runs are consolidated on a common (size+angle-filtered) real test set.

9.2 The Gap Is Real but Modest, and Partly a Quality Effect

A recurring theme is that the raw real-versus-synthetic comparison is misleading. Only after matching both runway size and vertical angle does the expected ordering appear (each model best on its own source; real no longer the easiest set), and even then the gap is modest. Much of the synthetic difficulty is an *image-quality* effect — poor runway/background separability, confusion with parallel structures, and softer edges on synthetic data — most pronounced for distant runways. The practical consequence is that cross-domain evaluations for this task must control for size and orientation, or they will systematically flatter the real set.

9.3 Dataset Diversity vs. Viewpoint Coverage

The **combined** source outperforms every individual source on real data despite covering roughly one fifth as many locations, indicating that *rendering-source diversity outweighs runway-viewpoint coverage* in this setup. Consistent with the LARD-V2 benchmark, no single source wins everywhere: satellite sources (**ges**, **arcgis**, **bingmaps**) lead under glare and nominal, while the 3D simulator **flsim** leads under rain, fog and night. The augmentation study confirms the same lesson from the other direction: appearance augmentation of a single source does not reach the diverse combined baseline.

9.4 Limitations

Several limitations qualify these results. The weather-augmentation sweep uses a reduced 20-epoch budget and, even with a fixed training set, four runs do not fully average out the sampling variance; larger effects are trustworthy, small ones less so. The weather-stratified real splits — especially fog and night — are small, so per-scenario numbers are noisier than the nominal figure. Self-training is exposed to confirmation bias, as the rain pseudo-label composition shows (quantity $\neq$ quality), and its largest gains are on in-distribution custom sequences rather than on the independent LARD set. Finally, the style-transfer stage and the tracker-augmented self-training variant are still ongoing, so the cross-technique comparison in Table 7 is not yet complete.

10 Conclusion and Future Work

10.1 Summary

This thesis studied the Sim2Real gap for vision-based runway detection with a YOLOv8-pose model fine-tuned on nominal-condition synthetic data from LARD-V2 and evaluated on a weather-stratified real test set from LARD-V1. Under a single, lightweight, reproducible protocol (fixed 5000-image budget, shared filtered test set, primary metric $\text{mAP}@[.50:.95]$), it first *quantified* the gap and then compared three strategies to *close* it.

The headline findings are: (RQ1) once the test sets are matched on both runway size and *vertical angle*, a modest but genuine Sim2Real gap appears — previously hidden because the real set looked deceptively easy — and much of the residual synthetic difficulty is an image-quality effect concentrated on distant runways; the combined source generalizes best overall, with different sources suiting different weather scenarios and fog the hardest. (RQ2) Weather augmentation gives limited and unstable gains, the scenario-matched augmentation is often not the best, and it does not substitute for source diversity. (RQ3) Self-training on unlabeled real landing sequences is the most effective technique so far, improving adverse LARD splits by over 10 mAP and in-distribution sequences by up to 35 mAP, though it is sensitive to pseudo-label noise. (RQ4) Control-guided diffusion style transfer, designed to import real appearance while preserving the corner labels, is ongoing.

10.2 Contributions Revisited

Against the LARD-V2 study, which analyzed cross-dataset generalization on synthetic test sets, this thesis contributes: a real-domain re-analysis of source suitability; a weather-stratified real benchmark extended with private landing sequences to compensate for sparse adverse-weather data; the vertical-angle filtering needed to measure the gap fairly; and a head-to-head comparison of weather augmentation, self-training and style transfer for lightweight runway-corner detection.

10.3 Future Work

Several threads remain open. The **tracker-augmented self-training** variant (Section 7.6) should be completed and compared against the confidence-only loop, quantifying the benefit of temporal consistency for distant runways. The **diffusion style-transfer** stage should be evaluated and, time permitting, contrasted with a CycleGAN baseline to measure both accuracy and label integrity. Beyond these, promising directions include *combining* the techniques (e.g. self-training on style-transferred data), extending the evaluation to full pose/geometry metrics, enlarging and balancing the adverse-weather real splits, and closing the far-runway fidelity gap identified in Section 5.5.

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A Additional Materials

Optional supporting material: full per-run augmentation tables, additional cross-source metrics, qualitative detection examples, style-transfer image samples, and dataset/statistics details.

A.1 Reproducibility

Software versions (Ultralytics/YOLOv8, albumentations, tracker, Cosmos Transfer 2.5), random seeds, and command lines/configuration files used for the runs.